



TEAM NAME: END OF BEGINNING

COLLEGE NAME: SRI VENKATESWARA COLLEGE OF ENGINEERING

PROBLEM STATEMENT: AI-POWERED JOB SCAM DETECTION FOR

INDIA'S MIGRANT WORKERS

PROBLEM STATEMENT

"Every day, approximately 500 migrant workers in India transfer their life savings to a phone number. By morning, the number is dead. The agent is gone. The debt remains — for years."

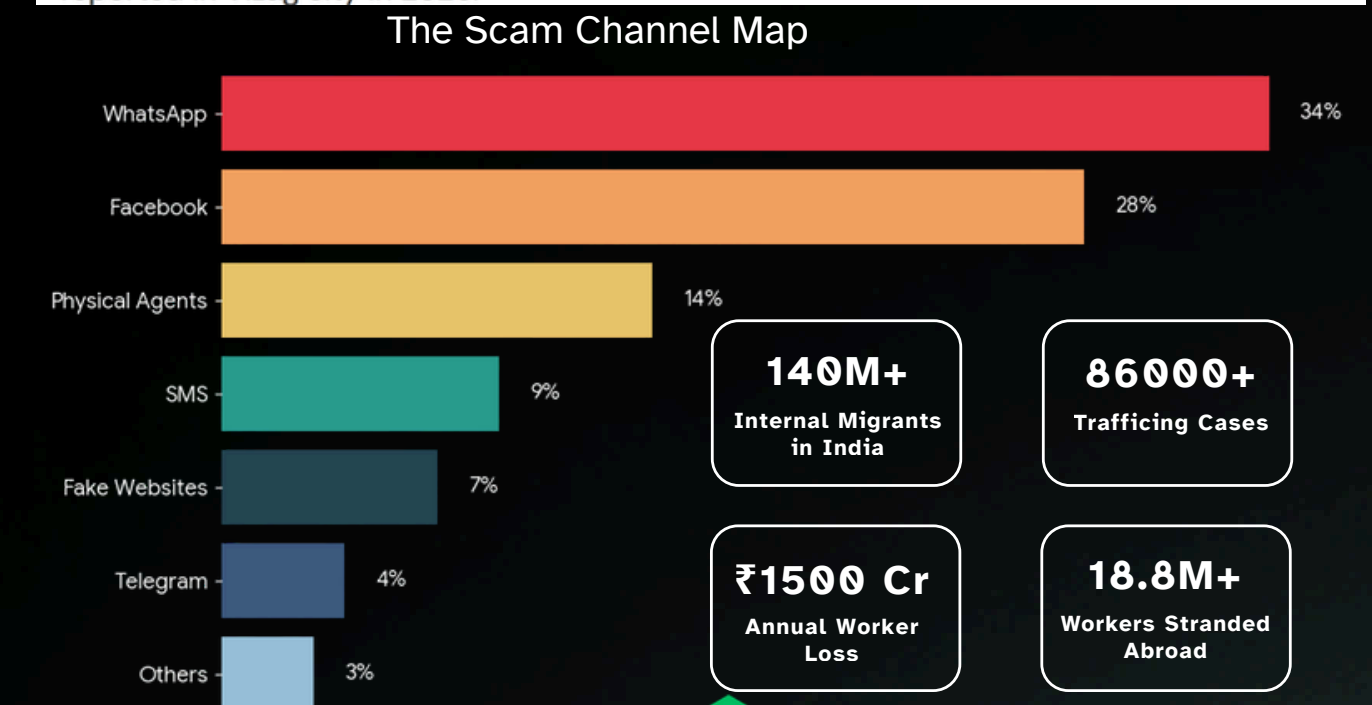
The Crisis: A National Emergency

- **The Victim:** 140M+ internal migrants, primarily semi-literate workers from states like Bihar and UP, who rely on WhatsApp as their sole digital lifeline.
- **The Hook:** Organized syndicates systematically target this demographic with fake job offers across WhatsApp, Facebook, and village agents.
- **The 24-Hour Trap:** Within one day of receiving a scam message, a worker typically borrows money at 36-60% annual interest to pay "registration fees" via UPI.
- **The Disappearance:** Once the money is sent, the agent vanishes and the phone number goes dead; the job never existed.



Current Solutions: Why They Fail

- **The Digital Gap:** Government tools like the eMigrate portal require English literacy and desktop access, making them inaccessible to the actual victims.
- **Delayed Response:** Official investigation cycles (FIR-to-arrest) take 8-14 months, by which time the money is long gone.
- **The Core Gap:** There is currently no real-time, vernacular tool that can verify a job message at the exact moment a worker is deciding whether to pay.



SOLUTION



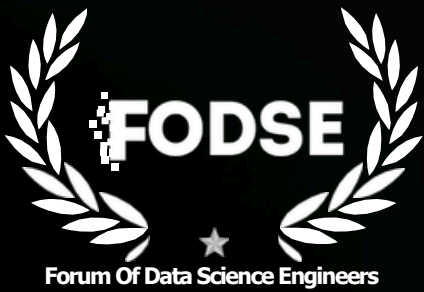
"ScamShield detects the scam before the worker sends the money."

Real-time · Multilingual · AI-powered · Free for workers

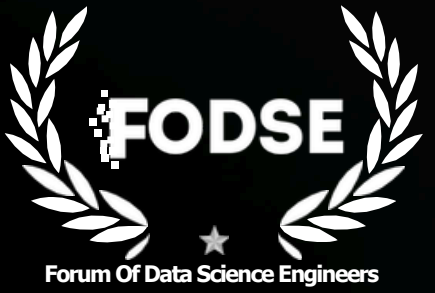
ScamShield — Your Job's Best Friend

- Forward any suspicious job message to the **WhatsApp bot** — get a Hindi risk verdict in under 30 seconds
- AI spots the red flags instantly — upfront fees, fake companies, salaries too good to be true
- **Browser extension** auto-badges every job post on Facebook & OLX as Safe or Danger — no clicking, no searching
- Upload a fake offer letter photo — system cross-checks it against MCA & government registries live
- Works on **voice notes** too — scam audio gets transcribed and analysed automatically
- **SMS shortcode** for feature phone users — no smartphone, no problem
- NGO field workers get an **offline mobile app** for bulk verification and victim tracking
- Government gets a **live dashboard** — fraud heatmaps, syndicate networks, one-click FIR export
- Zero cost to build. **Multilingual**, hence Zero English needed. Works at 10 PM when the scam arrives.

FLOW DIAGRAM



TECHNICAL REQUIREMENTS



The project uses Python with FastAPI for backend development, React.js for the frontend, PostgreSQL for database management, TensorFlow for machine learning-based fraud detection, OpenCV for document verification, Whisper for speech analysis, and Railway for cloud deployment. Messaging integration is achieved using WhatsApp Business API, while security is handled through JWT authentication.



Hugging Face



Whisper



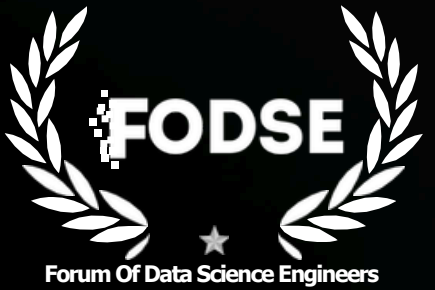
Railway

WHATSAPP
BUSINESS
API



Chart.js

NOVELTY / USE CASE



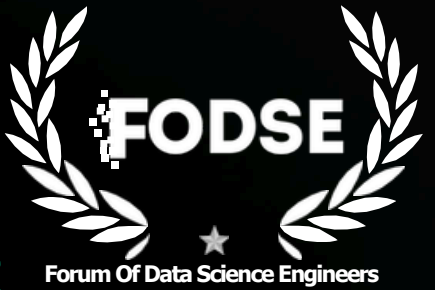
Technical Novelty

- Real-Time "Moment of Decision" Protection: Unlike retrospective NGO awareness camps, the AI provides a risk verdict in under 30 seconds, intervening before the worker transfers any money .
- Multilingual & Vernacular-First Engine: Native support for 12+ Indian languages (Hindi, Bhojpuri, Odia, Bengali, etc.) using AI4Bharat IndicTrans2, specifically designed for semi-literate users.
- Multimodal AI Intelligence: A unified backend that processes text (NLP), voice notes (OpenAI Whisper), and physical documents (Tesseract OCR) to detect diverse scam formats.
- Deterministic Government Cross-Checks: Goes beyond probabilistic AI by verifying extracted entities (GST, CIN, License numbers) against MCA21 and eMigrate RAPS registries
- Syndicate Network Graphing: Uses a graph database to map connections between phone numbers and UPI IDs across different victim reports to identify organized trafficking rings .

Real World Use Case

- Target Demographic: A migrant worker from a high-risk district (e.g., Deoria, UP) seeking overseas employment.
- The Scam Trigger: A fraudulent agent sends a WhatsApp forward promising high wages but demanding a ₹8,000 "registration fee" by midnight .
- The Action: Instead of borrowing from a moneylender at 36-60% interest, the worker forwards the message or a photo of the "offer letter" to the NaukariSaathi bot.
- The System Response: The AI detects a fake GST number and flags the contact as a reported scam. The bot instantly replies in the worker's mother tongue: "KHATRE KI BAAT! 89% probability this is fraud. Unregistered company. Do not pay any money".
- The Socio-Economic Impact: Directly prevents a 2-5 year debt spiral for the family and acts as a barrier against human trafficking recruitment pipelines.

BUSINESS MODEL / TARGET MARKET



KEY RESOURCES

- Multilingual AI classifier
- Scam intelligence database
- Govt API integrations
- WhatsApp Business API
- Dev team + NGO partners

CHANNELS

- WhatsApp & Telegram bots
- Browser extension
- SMS / IVR shortcode
- Field Worker PWA
- Admin dashboard

VALUE PROPOSITION

- Real-time scam detection
- Works on Rs.500 phones
- 30-second risk verdict
- Zero cost to workers
- Govt-grade data security

CUSTOMER RELATIONSHIPS

- WhatsApp self-service bot
- NGO field worker support
- Govt dashboard access
- Agency verified badge

CUSTOMER SEGMENTS

- 140M+ migrant workers
- Worker families
- NGO field counsellors
- State Labour Depts
- Cyber Crime Cells
- eMigrate agencies

COST STRUCTURE

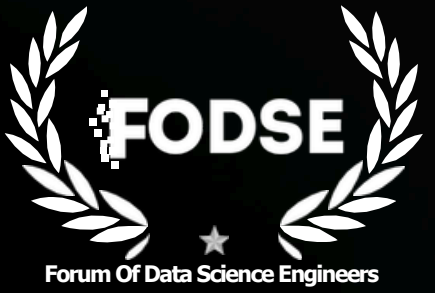
- Cloud hosting (Railway / Supabase / Vercel) — near zero on free tiers
- Twilio SMS / WhatsApp API — usage-based, low volume initially
- AI inference (Groq / Whisper local) — free tier sufficient for MVP
- Team salaries & NGO pilot coordination

REVENUE STREAMS

- Phase 1: CSR & govt grants — free to workers
- Phase 2: State intelligence reports — ₹50K–2L/state/month
- Phase 3: Verified Agency Badge — annual B2B subscription
- Phase 4: eSHRAM API integration — MoU-based govt contract



TEAM DETAILS



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